

Convolución de dos funciones

Definición 1 (Convolución). Sea (X, Σ, μ) un espacio de medida $f, g: X \rightarrow \mathbb{K} = \mathbb{R} \text{ o } \mathbb{C}$ dos funciones medibles, $f * g: X \rightarrow \mathbb{K}$ es la convolución de f con g

$$\iff \forall x \in X : f * g(x) := \int_X f(x - y) g(y) d\mu(y).$$

Observación 2. Si $f \in \mathcal{L}^\infty(\mathbb{R}^n)$ y $g \in \mathcal{L}^1(\mathbb{R}^n)$, entonces $f * g$ está bien definida c.t.p. y

$$|f * g(x)| \leq \int_{\mathbb{R}} |f(x - y)| |g(y)| dy \leq \|f\|_\infty \|g\|_1 \quad \text{en c.t.p. } x \in \mathbb{R}^n.$$

Referenciado en

- `lem-convolucion`
- `lem-transformada-fourier-convolucion`
- `obs-sumacion-cesaro-convolucion-nucleo-fejer`
- `prop-convolucion-exp-conjugados`
- `obs-sumas-parciales-serie-fourier-convolucion-nucleo-dirichlet`
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